Empirical Spectral Geometry of MEC Covariance Structure

(Dated: July 6, 2026)

High-dimensional dynamical systems under global stabilization exhibit a characteristic collapse of their effective dimensionality, measured by the participation ratio PR, toward PR ≈ 1. We analyze 21 recordings from rodent medial entorhinal cortex (MEC) and 22 synthetic dynamical conditions across five architectural classes. MEC covariance spectra follow a slow exponential decay with rate $\alpha = 0.039 \pm 0.018$ and participation ratio PR $= 37 \pm 20$, indicating broad but non-uniform spectral support. Synthetic stabilized systems cluster at $\alpha > 0.5$ and PR < 3. A reference curve PR $= (1 + e^{-\alpha})/(1 - e^{-\alpha})$, derived from a pure geometric series, provides a baseline for comparison: MEC spectra are systematically more concentrated than this reference, while collapsed synthetic systems cluster near it. The broad-spectrum organization resides in instantaneous correlation geometry rather than temporal propagation. These results establish a reproducible empirical characterization of spectral organization in MEC dynamics and identify the gap between biological correlation geometry and synthetic nulls as a target for mechanistic modeling.

I. INTRODUCTION

High-dimensional dynamical systems—from neural populations to climate models to complex fluids—face a fundamental tension between stability and representational richness [1, 2]. Maintaining many active degrees of freedom while preventing instability is a generic challenge across domains.

Previous work has identified specific collapse mechanisms: the dazzling problem in neural networks [3], mode collapse in generative models [4], and spectral compression in recurrent networks [5]. What remains unclear is whether these collapse phenomena share common spectral signatures across implementations.

Here we address this question by analyzing 43 dynamical conditions: 21 MEC recordings and 22 synthetic systems. We characterize their spectral geometry using the covariance matrix of instantaneous activity and find a reproducible separation between biological and synthetic systems in the space defined by spectral decay rate α and participation ratio PR. The primary contribution is empirical and descriptive: showing that MEC occupies a distinct spectral regime that is reproducible across recordings, robust to preprocessing, and separable from the collapsed regime accessible to synthetic stabilizing mechanisms.

II. DEFINITIONS

We analyze the covariance matrix $\Sigma = \langle \mathbf{x}\mathbf{x}^\top \rangle_t$ of each trajectory $\mathbf{x} \in \mathbb{R}^N$, with z-scored activity (zero mean, unit variance).

a. Participation Ratio. Given eigenvalues $\lambda_1 \geq \cdots \geq \lambda_N$ of Σ, the participation ratio

$$\mathrm{PR} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \tag{1}$$

measures the effective number of dimensions carrying signal [6]. PR ranges from 1 (single active dimension) to N (uniform variance).

b. Spectral Decay Rate. The spectral decay rate α is defined by the exponential fit

$$\lambda_k \sim e^{-\alpha k}, \tag{2}$$

estimated from eigenvalues 2 through $N/2$ via least-squares regression on log-transformed values. This range excludes the leading eigenvalue (often an outlier) and the tail (where finite-size noise dominates). Small $\alpha \ll 0.1$ indicates broad spectral support; large $\alpha \gg 0.5$ indicates rapid decay and effective low-dimensionality.

A pure geometric series $\lambda_k = \lambda_1 \cdot e^{-\alpha(k-1)}$ produces

$$\mathrm{PR}_{\mathrm{geom}}(\alpha) = \frac{1 + e^{-\alpha}}{1 - e^{-\alpha}}, \tag{3}$$

which serves as a reference curve for comparison with empirical spectra.

c. Manifold Closure. The closure metric LC quantifies reconstruction error of a linear projection:

$$\mathrm{LC} = \log_{10} \frac{\|X - X_{\mathrm{proj}}\|_F^2}{\|X\|_F^2}, \tag{4}$$

where X_{proj} retains the components needed to explain 95% of variance. More negative LC indicates stronger trajectory collapse.

III. EXPERIMENTAL ARCHITECTURE

We analyzed 43 dynamical conditions (Table I), comprising 21 biological recordings from rodent medial entorhinal cortex (MEC) during spatial navigation and 22 synthetic conditions.

a. Biological recordings. The MEC dataset includes 21 recording sessions from 4 rodent cohorts (Calais, Goa, Kerala, Mumbai), each containing 30–120 seconds of spiking activity from 100–600 neurons during open-field foraging. Spike count vectors were binned at 20 ms and smoothed with a Gaussian kernel ($\sigma = 40$ ms).

b. Synthetic systems. We constructed five architectural classes: (i) reservoir computers with random recurrent connectivity [7, 8], (ii) additive field dynamics, (iii) figures/fig1_collapse_law.pdf

FIG. 1. **Empirical collapse pattern.** Participation ratio PR vs spectral decay rate α for all 43 conditions. MEC recordings (blue) cluster at low α and high PR; synthetic systems (orange) collapse to $\mathrm{PR} \approx 1$ at high α. The dashed curve shows the geometric-series reference $\mathrm{PR} = (1+e^{-\alpha})/(1-e^{-\alpha})$ (Eq. 3). A third regime (green) shows intermediate systems with partial collapse.

figures/fig2_convex_void.pdf

FIG. 2. **Empirical gap in reachable dynamics.** The (PR, LC) plane showing the convex hull of all tested conditions (shaded). An unpopulated region exists at $\mathrm{PR} < 15$, $\mathrm{LC} < -5$: no tested system produces simultaneously low dimensionality and strong collapse. The dashed line marks the empirical boundary $\mathrm{LC} = -2\log_{10}(\mathrm{PR}) + 0.6$.

wave field dynamics, (iv) latent manifold models, and (v) constrained random walks. Each class was run with and without global stabilization operators: homeostatic gain control, additive transport currents, and anisotropic suppression [9, 10].

IV. EMPIRICAL COLLAPSE PATTERN

Figure 1 shows the central empirical finding: participation ratio as a function of spectral decay rate across all 43 conditions.

Two distinct regimes emerge. MEC recordings occupy $\alpha < 0.1$ and $\mathrm{PR} > 10$, with mean $\alpha = 0.039 \pm 0.018$ and $\mathrm{PR} = 37 \pm 20$. Synthetic systems cluster at $\alpha > 0.5$ and $\mathrm{PR} < 3$, indicating effective one-dimensional dynamics. A third group of intermediate systems bridges these regimes.

The geometric-series reference curve (Eq. 3) provides a baseline. Synthetic collapsed systems cluster near this curve, while MEC systems systematically fall below it: for their α, they have lower PR than a pure geometric series predicts. This indicates that MEC spectra are more concentrated in the leading eigenvalues than a simple exponential decay would produce—the tail decays faster than the exponential fit to the leading half suggests.

A second empirical relation connects PR and LC:

$$\mathrm{LC} \leq -2\log_{10}(\mathrm{PR}) + 0.6. \tag{5}$$

Systems with broad spectral support (high PR) maintain low closure (negative LC), while collapsed systems exhibit LC near zero.

V. EMPIRICAL SPECTRAL GAP

The empirical regularities define a region in the (PR, LC) plane that remains unpopulated. Figure 2 shows the convex hull of all tested conditions.

The reachable region is bounded by three empirical frontiers representing tradeoffs between spectral breadth, dimensional collapse, and trajectory closure. An empirical gap exists at low PR and highly negative LC—states combining low dimensionality with strong collapse are dynamically inaccessible under the tested conditions. This gap is observed across all architectural classes.

VI. EMPIRICAL RELATIONS

The product $\alpha \cdot \mathrm{PR}$ separates dynamical classes (Fig. 3). For MEC, $\alpha \cdot \mathrm{PR} \approx 1.4 \pm 0.5$, reflecting a consistent tradeoff between spectral decay rate and effective

TABLE I. Summary of dynamical conditions analyzed.

Category	Conditions	N range	Stabilization types
MEC recordings	21	100–600	Intrinsic
Reservoir computer	8	100–500	Homeostatic, transport, anisotropic
Additive field	6	50–200	Transport, anisotropic
Wave field	4	100–300	Homeostatic, transport
Latent manifold	2	100	Homeostatic
Constrained random walk	2	50–100	None

figures/fig3_conservation.pdf

FIG. 3. **Empirical spectral relations.** $\alpha \cdot \mathrm{PR}$ as a function of α for all conditions. MEC (blue) clusters near $\alpha \cdot \mathrm{PR} \approx 1.4$. Synthetic systems (orange) show large $\alpha \cdot \mathrm{PR}$. The intermediate group (green) bridges the regimes.

figures/fig4_phase_diagram.pdf

FIG. 4. **Regime classification.** Distribution of all conditions in (α, PR, LC) space. The broad-spectrum zone ($\alpha < 0.1$, PR > 15, LC < -5) contains 15 of 21 MEC recordings and 0 of 22 synthetic conditions. The collapse zone ($\alpha > 0.5$, PR < 3) contains 18 of 22 synthetic conditions and 0 MEC recordings.

dimensionality. For synthetic collapsed systems, $\alpha \cdot \mathrm{PR}$ is large as spectral decay dominates.

A second empirical relation involves the product $\mathrm{PR} \cdot \exp(\mathrm{LC}/2) = 5.8 \pm 1.0$ for MEC systems, independent of recording session.

VII. REGIME CLASSIFICATION

The three metrics (α, PR, LC) define distinct dynamical regimes. Figure 4 maps all 43 conditions in this space.

The broad-spectrum zone, defined by $\alpha < 0.1$, PR > 15, LC < -5, contains 15 of 21 MEC recordings and none of the synthetic conditions. The collapse zone, defined by $\alpha > 0.5$, PR < 3, contains 18 of 22 synthetic conditions and no MEC recordings. The remaining conditions occupy a transition zone.

VIII. STABILIZATION OPERATOR COMPARISON

To test whether the collapse pattern depends on the specific stabilization mechanism, we applied three operators—homeostatic gain control, additive transport, and anisotropic suppression—to a reservoir computer architecture. Figure 5 shows the results.

All three operators drive systems toward collapse along similar spectral trajectories. The operator affects the rate and path of collapse but not its ultimate regime in (α, PR) space. This suggests the collapse pattern reflects a property common to stabilized high-dimensional dynamics rather than a mechanism-specific artifact.

figures/fig5_operators.pdf

FIG. 5. **Stabilization operator comparison.** PR and α for reservoir computers with different stabilization operators. All operators drive the system toward collapse (increasing α, decreasing PR), following a similar spectral trajectory. Operator choice affects the rate of collapse but not its ultimate destination.

IX. DISCUSSION

We have characterized the spectral geometry of dimensional collapse in stabilized high-dimensional dynamical systems. The central empirical result—a reproducible separation between MEC recordings and synthetic systems in (α, PR) space—holds across 43 conditions spanning biological and synthetic architectures.

a. Geometric vs mechanistic description. The observed spectral patterns are properties of the instantaneous correlation geometry. The precision matrix, estimated via graphical lasso [11, 12], encodes a sparse ($\approx 78\%$ zeros) high-dimensional (PR ≈ 98) interaction network [13]. Temporal operators produce no comparable structure, indicating that the organization is geometric rather than propagative. The spectral properties of the precision graph suggest distributed rather than modular architecture [14, 15].

b. Relationship to random matrix theory. The reference curve Eq. 3 parallels eigenvalue bounds in random matrix theory [16, 17], but the spectra studied here deviate systematically from classical ensembles [18, 19]. The systematic deviation of MEC spectra below the geometric-series reference is quantified in the companion paper: MEC eigenvector statistics differ from GOE at 3.47σ and from sparse $p = 0.05$ ensembles at 0.86σ for fixed $N \approx 120$, with the latter resolved by scaling separation $d(N) \sim N^{2.2}$.

c. Open questions. The origin of the broad-spectrum regime ($\alpha < 0.1$, PR > 15, LC < -5) observed in MEC recordings remains unresolved. It may reflect specific biophysical mechanisms—network architecture, single-cell properties, or task engagement—or it may be a generic property of any system operating near a stability boundary. The fact that no synthetic stabilization operator reproduced this regime suggests a limitation of the current operator set rather than an impossibility result. Whether this regime reflects a distinct organizational principle or a finite-size effect observable only in large-N systems remains an open question [20, 21].

d. Limitations. This study is limited to $N \lesssim 600$ dimensions. Finite-size scaling analysis [22, 23] (companion paper) shows the separation between MEC and sparse null ensembles grows with system size as $d(N) \sim N^{2.2}$, consistent with genuine structural organization rather than finite-size artifacts. However, the N range does not permit definitive extrapolation to asymptotically large systems [24]. The analysis uses linearized spectral measures; nonlinear dynamics may introduce additional structure beyond the covariance spectrum. The biological recordings are from a single species and brain region [25, 26]; generalization requires further validation [27].

Because recordings are reused across subsampling analyses, effective sample counts for scaling estimates may be lower than nominal bootstrap counts. The null-model family tested here—Poisson, shuffled, and sparse ensembles—does not exhaust the space of possible nulls; additional null constructions (e.g., state-space models, latent factor models) may produce different separation profiles. All preprocessing choices (20 ms binning, $\sigma = 40$ ms smoothing, z-scoring) affect spectral estimates; the robustness of our conclusions across this pipeline is documented in the companion paper but cannot be taken as proof of invariance under arbitrary preprocessing. The claims in this paper are observational and descriptive: we characterize the spectral geometry of MEC covariance structure and its separation from tested nulls, but do not identify the mechanism producing this organization.

DATA AVAILABILITY

All code, data, and reproduction scripts are archived at doi:10.5281/zenodo.21223156. The repository includes the full preprocessing pipeline, statistical comparison module, and all null-control experiments.

ACKNOWLEDGMENTS

We thank the members of the SGP collaboration for discussions and comments.

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